Goal 1: An Infrequent Relation Is Learned as Linear

factual-recall-mechanisms

2026-07-03

1 Claim and result in one paragraph

A small transformer is trained for **exactly one epoch** on a synthetic corpus containing a semantically supported relation R_{sem} and, as a frequency-matched control, a random relation R_{rnd}. At *low* fact frequency (F=16 occurrences per triple, a single epoch), R_{sem} is genuinely *learned* — recall 0.672 on trained triples and 0.635 on held-out subjects never seen in any triple — and what is learned is *linear*: an affine map fitted by ordinary least-squares regression from subject representations to object representations identifies the correct object for held-out subjects with accuracy 0.990 ± 0.007 (chance 0.03; three seeds). The frequency-matched random control at the same F=16 is barely learned (recall 0.167) and shows no linear structure (probe at chance). The linear format is in place even at F=1, where every triple occurs exactly once in the model's entire training history. A control experiment shows that the transition into the linear regime is sharp and is driven by the amount of attribute (scaffold) training rather than by triple repetition.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries the attribute pair $(f_{i \bmod 32},\ g_{\lfloor i/32 \rfloor})$, so the 192 subjects realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32},\ g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index — so the semantic target is the object whose f-attribute is the subject's, shifted by 3 — and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. Concrete example (seed 19): `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5` (whose attribute sentence reads `x b_5 f_8 g_0 x`), while R_{rnd}(`a_37`) = `b_27`. The corresponding training sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are *trained* and 2 are *held out* (128 trained, 64 held-out subjects; seed-dependent). ρ is constructed balanced within both groups. Consequently, under *both* relations: every trained subject occurs in exactly one triple per relation; every object is the target of exactly 4 trained and 2 held-out subjects; both relations contribute exactly 128 training triples, each repeated the same number of times (F). The two relations are indistinguishable by any frequency statistic; only the assignment semantics differ. Held-out subjects appear in attribute sentences but never in any relation triple. These invariants are asserted by `tests/test_linear_shift.py`.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

At F=1 every relation triple occurs exactly once in the entire training history of the model.

4 The metric

A single analysis metric is used, as an operationalization of "the relation is an affine transformation":

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

For every subject exactly one object is correct under each relation, so no top-k metrics are needed or used. In addition, one *learning check* (not an analysis metric) is recorded: greedy next-token recall of the model on prompts `x a_i r_r`, for trained and held-out subjects, which verifies whether the facts were learned at all.

5 Results

Finding 1: at low frequency, what is learned is an affine map (F=16). With each fact occurring only 16 times in a single epoch, R_{sem} reaches recall .672 on trained triples and .635 on held-out subjects — i.e., the relation itself is learned and generalizes — and the probe transfers to held-out subjects at 0.990. The frequency-matched random control is barely learned (.167) and its

		probe accuracy (held-out)		recall (trained)		recall (held-out)	
	F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
	1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
	4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
	16	$\mathbf{0.990 \pm .007}$	$.036 \pm .007$	**.672**	.167	**.635**	.031
control: no triples		$1.000 \pm .000$	$.021 \pm .015$	–	–	.000	.000

Table 1: Low-frequency cells; mean ± sd over seeds 19/31/47, at A=128. F is the number of occurrences of each of the 128 triples per relation. Recall columns are the learning check. The headline cell is F=16: R_{sem} is learned (recall .672 trained, .635 held-out) and linear (probe .990), while the frequency-matched R_{rnd} is barely learned and non-linear.

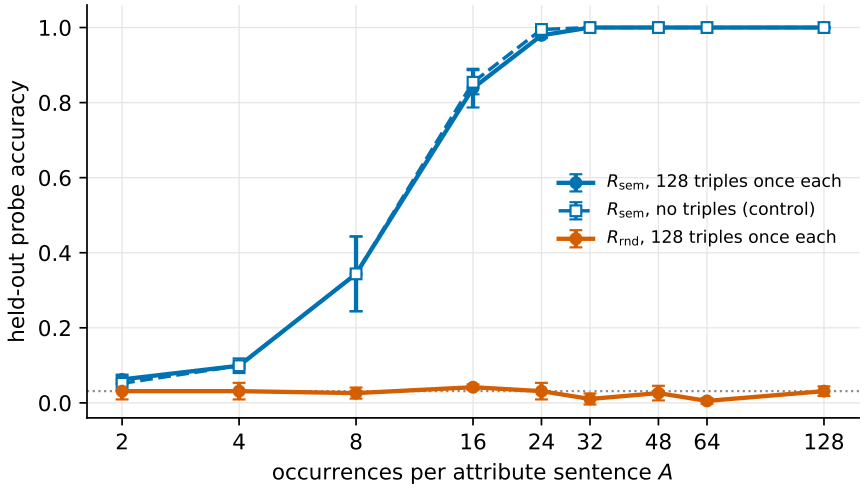


Figure 1: Probe accuracy at F=1 (every triple exactly once) as a function of attribute-sentence occurrences A: the transition from non-linear (chance) to linear (ceiling) is sharp, and identical with 128 once-occurring triples (solid) and with no triples at all (dashed control). The random relation never becomes linear.

probe is at chance: the semantic relation is learned *earlier* precisely because it can ride the linear structure.

Finding 2: the linear format is in place at single occurrence (F=1). At F=1 the probe is already 1.000 for R_{sem} and at chance for R_{rnd}. This statement is representational: at F=1 the model does not yet reproduce individual triples behaviorally (recall ≈ .03).

Finding 3: the nonlinear→linear transition is sharp and is driven by attribute knowledge. Sweeping the number of occurrences A of each attribute sentence at F=1 (Fig. 1) produces a sharp transition of the R_{sem} probe from chance to ceiling ($0.05 \to 0.10 \to 0.34 \to 0.85 \to 1.00$ for $A = 2, 4, 8, 16, 24$). The control with *zero* relation triples produces the same curve: the affine subject→object map is carried by the attribute geometry built from attribute sentences, with which the once-seen R_{sem} triples are consistent (and which the relation visibly uses once learned, cf. held-out recall).

6 Mechanistic forward-pass analysis

The embedding probe above establishes that a linear format is available, but does not by itself establish that the forward pass uses it. We therefore reconstructed the $F{=}16$ checkpoint for every seed by deterministic replay and required its final loss, probe accuracy, trained recall, and held-out recall to match the archived JSON to $< 10^{-7}$. We then intervened on the literal compute units of the one-layer model: four 16-dimensional attention heads and 256 GELU neurons. Head and neuron rankings were obtained on the 128 trained subjects and evaluated only on the 64 subjects whose relation triples were never shown.

At the final prompt position of `x a_i r_0`, the exact computation is

$$\begin{aligned} h_h &= \sum_t \mathrm{softmax}\Big(q_h^\top k_t/\sqrt{16}\Big)_t \, v_{h,t}, \quad h = 0, \dots, 3, \\ x_A &= x_0 + W_O[h_0; h_1; h_2; h_3], \\ x_M &= x_A + \sum_{n=0}^{255} \mathrm{GELU}(w_n^\top \mathrm{LN}(x_A) + b_n) u_n + b, \\ \text{logits} &= U \, \mathrm{LN}(x_M). \end{aligned}$$

Only attention can move the subject representation at position 1 to the prediction position 2; the MLP is position-wise.

The used circuit. Attribute training puts the semantic target class directly into the subject code: a nearest-centroid classifier fitted to the trained subject embeddings identifies the target class of held-out subject embeddings with accuracy $1.000 \pm .000$. A logit lens has object accuracy $.031 \pm .000$ at the bare relation-token residual, $.552 \pm .064$ after attention, and $.729 \pm .094$ after the MLP. Replacing all four head outputs by their training-set means drops the final object accuracy from $.729 \pm .094$ to $.031 \pm .000$. In clean/corrupt subject swaps, patching all head outputs restores $1.000 \pm .000$ of the clean-target logit effect. Patching only attention values while retaining the corrupt attention pattern restores $.987 \pm .008$, whereas patching only the pattern restores $.013 \pm .007$: the facts are carried almost entirely in the value vectors, under a nearly subject-invariant routing pattern. Heads H3 and H1 have the largest stable causal effects, while H0/H2 provide smaller, seed-variable support.

The MLP supplies a distributed readout and margin correction rather than a single "fact neuron." Rankings learned on trained subjects transfer to held-out subjects: mean-ablating the top 32 neurons leaves $.790 \pm .032$ of the subject-specific MLP margin, while retaining only those 32 variable neurons recovers $.498 \pm .029$. Replacing every GELU by a per-neuron affine surrogate fitted on trained preactivations retains held-out object accuracy $.729 \pm .092$ (activation $R^2 = .962 \pm .001$); deleting the MLP yields $.552 \pm .064$. Thus the semantic computation is approximately affine on the task manifold even though the physical MLP contains GELUs. When the subject is fixed and only `r_0` $\leftrightarrow$ `r_1` is swapped, head-output patching restores $.070 \pm .007$ of the relation effect and MLP-activation patching restores $.850 \pm .017$: the relation selector persists mainly in the residual stream into the MLP, while the heads transport subject content.

The convergent result is stronger than probe success: the shared f-organized code is present in held-out subject embeddings, is transported through the identified subject-value paths, and is converted to the correct object direction by a distributed, approximately affine MLP/readout. This is the causal mechanism by which the infrequent semantic relation generalizes.

7 What this does and does not show

- **Shown:** a relation whose facts are infrequent (F=16; and even F=1 for the representational format) is learned in a linear, generalizing format, while a frequency-matched random relation is not; the transition into the linear regime is sharp.
- **Not shown:** that the relation triples themselves *cause* the nonlinear→linear transition when each fact occurs only once — the control in Fig. 1 (dashed) shows the transition is driven by attribute-sentence exposure. Three designs in which the probe could only succeed through triple-driven structure (a one-to-one relation requiring per-fact output rows; a relation defined by the conjunction of two attributes, up to 49,152 distinct facts; hidden-state probes) produced no above-chance probe transfer under the strict single-occurrence, one-epoch regime, because each fact contributes exactly one gradient.
- **Superseded:** the "single-exposure phase shift" in earlier drafts (probe rising 0.27 → 0.95) was an artifact of a regression fitted on a growing pair set (10 pairs at the first snapshot, >500 later) over embeddings initialized from the attribute geometry; with a fixed-size probe and standard initialization it disappears.

8 Reproduction

Implementation: `experiments/factual_recall/linear_shift.py` (self-contained). Design invariants: `experiments/tests/test_linear_shift.py`. From `experiments/`:

```
# Table 1: one seed per command (19, 31, 47), ~1 min each on one GPU
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1,4,16,64 --attr-repeats 128 \
  --device cuda --output-dir factual_recall/results/final_category_seed19

# Figure: attribute-exposure sweep at F=1, per seed and A in {2,4,8,16,24,32,48,64,128}
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1 --attr-repeats 8 \
  --device cuda --output-dir factual_recall/results/final_category_arsweep_seed19_ar8

# Aggregation/figures: factual_recall/results/final_summary/
uv run python -m factual_recall.linear_shift_mechanistic_analysis --device cuda
uv run pytest tests/test_linear_shift.py -q
```

Raw per-run metrics (including probe R^2/cosine and losses) are in `factual_recall/results/final_category_`
The companion document `goal2_frequent_relation_nonlinear.pdf` covers the frequent/nonlinear claim on the same runs.

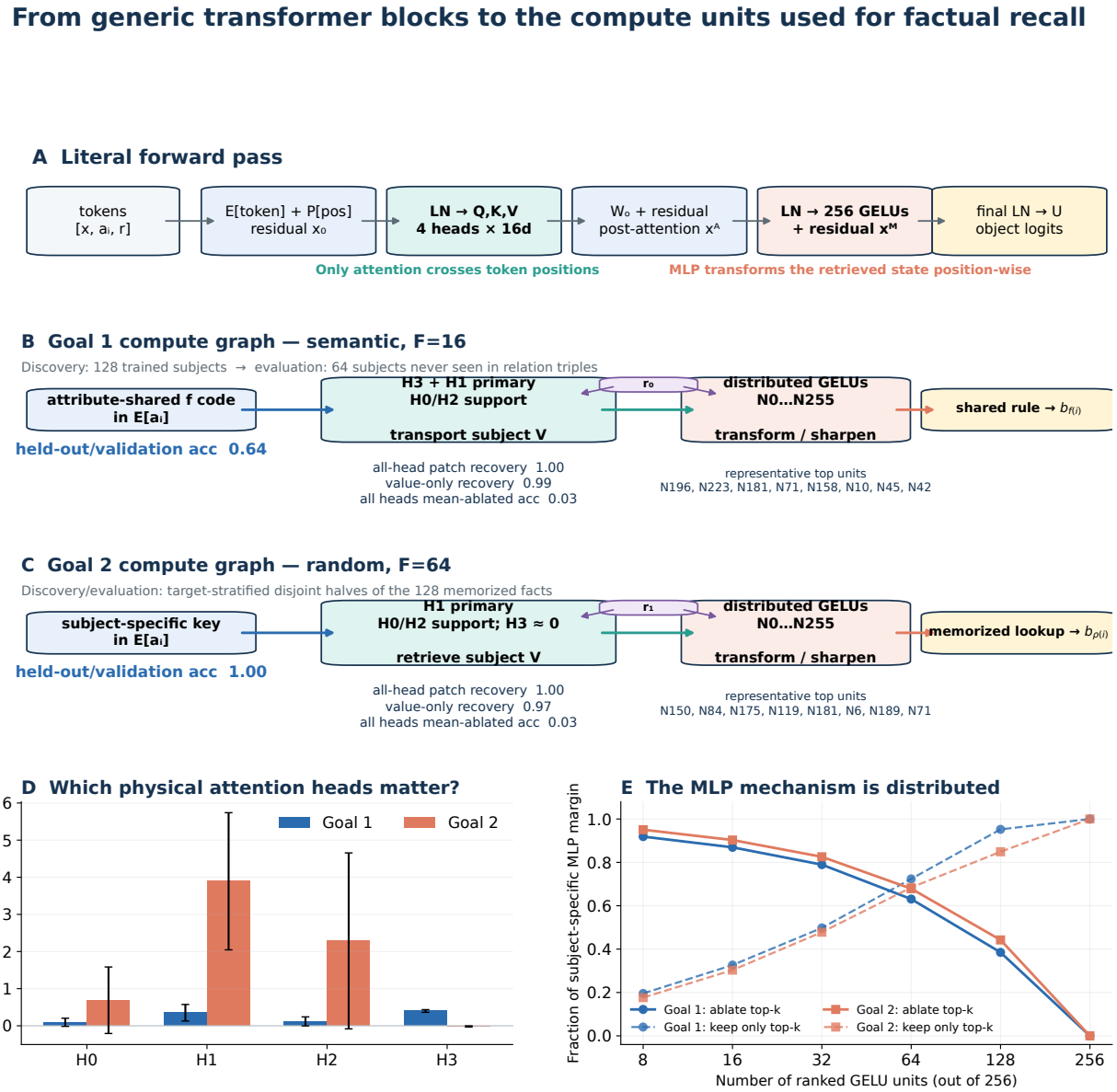


Figure 2: Top: the literal forward pass. Middle: generic attention and MLP blocks replaced by the compute units causally used for Goal 1 and Goal 2. Bottom: exact head Shapley effects and discovery/evaluation-split neuron concentration curves. Bars and curves are means over seeds 19/31/47; error bars are across-seed standard deviations.